Service Bulletin

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Lubricating Oil Consumption and Acceptable Limits	

Lubricating Oil Consumption and Acceptable Limits

This publication was written to familiarize field personnel with the causes of excessive lubricating oil consumption. Inspection and analysis of the internal engine components will provide a knowledge of the factors that contribute to piston ring and cylinder liner wear. Knowledge of the relationship between engine operation and the wear factors that cause excessive lubricating oil consumption will enable efforts to be directed toward extending the service life of the engine.

Excessive lubricating oil consumption is defined as the addition of lubricating oil to the crankcase at an unacceptably high rate.

Table 1, Acceptable Lubricating Oil Usage - Any Time During Coverage Period

Engine Family	Miles Per Quart	Miles Per Liter	Miles Per Imperial Quart	KM Per Quart	KM Per Liter	KM Per Imperial Quart	Hrs Per Quart	Hrs Per Liter	Hrs Per Imperial Quart
A-Series							10	10.6	12
B3.3/4B	400	425	475	650	675	775	10	10.6	12
ISF/QSF/F	400	425	475	650	675	775	10	10.6	12
ISV	400	425	475	650	675	775	10	10.6	12
6B/ISB/QSB	Reference the oil consumption evaluator under the "Service Tools" tab in QuickServe® Online.								
ISB6.7G/B6.7N/B6.7O	400	425	475	650	675	775	10	10.6	12
6C/ISC/QSC/ISL	400	425	475	650	675	775	10	10.6	12

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Engine Family	**Miles Per Quart**	**Miles Per Liter**	**Miles Per Imperial Quart**	**KM Per Quart**	**KM Per Liter**	**KM Per Imperial Quart**	**Hrs Per Quart**	**Hrs Per Liter**	**Hrs Per Imperial Quart**
ISL G/L9N	400	425	475	650	675	775	10	10.6	12
V/VT-378	250	265	310	400	425	485	4	4.3	4.8
V/VT-504	250	265	310	400	425	485	4	4.3	4.8
VT-555	250	265	310	400	425	485	4	4.3	4.8
L10	500	530	620	800	850	970	7	7.4	8.4
M11/ISM	500	530	620	800	850	970	7	7.4	8.4
ISG11	500	530	620	800	850	970	7	7.4	8.4
ISG/QSG12	500	530	620	800	850	970	7	7.4	8.4
ISZ/QSZ	500	530	620	800	850	970	7	7.4	8.4
N14/NT	500	530	620	800	850	970	7	7.4	8.4
ISX/QSX/Signature™	Reference the oil consumption evaluator under the "Service Tools" tab in QuickServe® Online.								
ISX12 G/ISX12N	500	530	620	800	850	970	7	7.4	8.4
V/VT/VTA-903	250	265	310	400	425	485	4	4.3	4.8
KT/KTA19	200	210	250	320	340	390	3	3.2	3.6
V/VT/VTA28							2	2.1	1.1
KT/KTA38							1.5	1.6	1.8
KTA50							1.1	1.2	1.3

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Engine Family	**Miles Per Quart**	**Miles Per Liter**	**Miles Per Imperial Quart**	**KM Per Quart**	**KM Per Liter**	**KM Per Imperial Quart**	**Hrs Per Quart**	**Hrs Per Liter**	**Hrs Per Imperial Quart**
QSK19							3	3.2	3.6
QST30							1.7	1.8	2
QSK23							1.7	1.8	2
QSK38							1.3	1.4	1.5
QSK45							1.25	1.3	1.5
QSK50							1	1.1	1.2
QSK60							0.9	0.95	1.1
QSK78							0.6	0.65	0.72
QSK95							0.55	0.58	0.66
HSK78							0.6	0.65	0.72

Cummins Inc. defines "Acceptable Lubricating Oil Usage" as outlined in the following table:

Table 2, Acceptable Lubricating Oil Usage (Transit Bus, Shuttle Bus, and School Bus) - Any Time During Coverage Period									
Engine Family	**Miles Per Quart**	**Miles Per Liter**	**Miles Per Imperial Quart**	**KM Per Quart**	**KM Per Liter**	**KM Per Imperial Quart**	**Hrs Per Quart**	**Hrs Per Liter**	**Hrs Per Imperial Quart**
B	Reference the oil consumption evaluator under the "Service Tools" tab in QuickServe® Online								
C	150	160	180	240	255	290	8.0	8.5	10.0
L, M, N	100	105	120	160	170	195	4.0	4.3	5.0
ISB6.7 G/ B6.7N /B6.7 O	400	425	475	650	675	775	10	10.6	12
ISL G/L9N	400	425	475	650	675	775	10	10.6	12

Categorizing/Generalizing

Because of “Hardware” improvements, a lubricating oil consumption complaint is a problem that **must** be evaluated on an engine-to-engine basis and **not** categorized or generalized.

Cylinder liner conditions which result in lubricating oil consumption are:

- Worn liners due to abrasives
- Scoring
- Scuffing.

Prior to Disassembly

Accurately determine the primary contributor(s) to the lubricating oil usage problem. An inaccurate analysis can result in unnecessary and costly repairs which may or may **not** solve the problem.

To evaluate a lubricating oil consumption complaint requires:

1. Analysis of the engine application
2. Review of the maintenance history
3. Inspection for external leaks.

Application Review

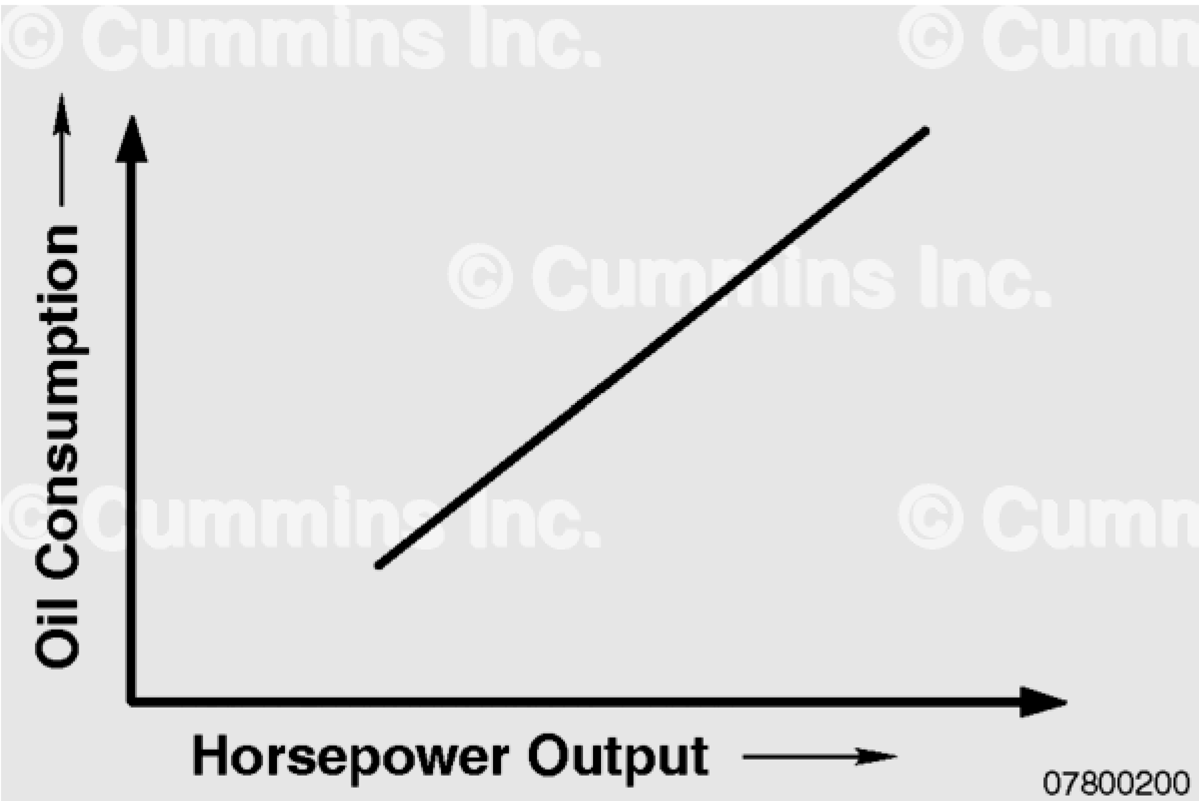

Figure 1: Lubricating Oil Consumption Rate

Application inspection is important because lubricating oil consumption depends strongly on horsepower output (load factor). (Figure 1). It should be determined if the engine is a 290 BHP engine in a 400 BHP engine application. If so, shorter life from a deterioration in lubricating oil control **must** be expected. To determine the average load factor on the engine, the engine rating, transmission information, rear axle ratio, route and gross vehicle weight outbound, and route and gross vehicle weight inbound are needed. Other useful application information is the period of engine idling during the winter.

From a failure analysis standpoint, determine if the engine lubricating oil level is correct. Is the dipstick or the automatic lubricating oil addition unit properly calibrated? Too much lubricating oil in the crankcase increases lubricating oil consumption. If extremely high, the crankshaft will dip into the lubricating oil and flood the cylinders and/or heat the lubricating oil which decreases viscosity.

Note the engine lubricating oil level on the dipstick, drain, and collect the lubricating oil from the engine. Measure the volume of lubricating oil collected. Was the dipstick calibration correct? If an automatic lubricating oil addition unit is used, was it correctly calibrated to maintain the lubricating oil level midway between the high and low levels? If the engine lubricating oil level was too high, correct the calibration of the dipstick or the automatic addition sensor.

Is the engine operating too hot or too cold? Overheating causes increased piston and lubricating oil temperature which result in faster lubricating oil degradation, and faster accumulation of piston carbon deposits. As a rule of thumb, for every 11°C [20°F] increase in temperature, the chemical reaction rates which produce deposits double.

Overcooling can cause condensation to form in the lubricating oil, thus causing the lubricating oil additive package to drop out of solution and possibility plug filters. The condensation occurs through the normal combustion process and will **not** fully vaporize and exit through the breather if overcooled.

Before engine disassembly, the air intake system should be completely and thoroughly checked. Look for any leaks on the clean side of the air cleaner and for any new components in the air intake system. Leaks or new components in the air intake system may indicate inadequate air filtration and subsequent abrasive wear of cylinder components. The air intake system should be checked by an appropriate method. Careful attention should be used in inspecting all plumbing connections and clamps. The gasket surface and baseplate flatness of some frontal air cleaner systems deserve special attention. This baseplate is easily warped, which results in problems with the air cleaner gasket sealing. Remember, most abrasive wear problems with cylinder components result from inadequate air filtration.

When identified, these application problems **must** be corrected. Correcting them will help alleviate a lubricating oil consumption problem or may prevent a repeat malfunction.

Maintenance History Review

A maintenance history inspection is important because it may provide clues for identifying contributors to the lubricating oil usage problem. A maintenance history inspection includes determining if maintenance was **always** performed by qualified personnel in shops with good shop practices. In all cases, and especially if the answer to the above question is NO, care should be exercised in examining the engine. Previous engine malfunctions, particularly coolant or fuel in the lubricating oil, and overheating should be noted. Coolant in the lubricating oil can result in lubricating oil additive drop-out and filter plugging. Loss of the lubricating oil additives results in increased wear, and inability of the lubricating oil to neutralize acidic combustion by-products. These acidic combustion products greatly increase the accumulation rate of piston deposits. Filter plugging results in the circulation of unfiltered lubricating oil, which may contain wear debris and dirt which accelerates cylinder component wear.

Fuel in the lubricating oil is known to increase the accumulation rate of piston deposits. This occurs because the volatile fuel migrates into the ring-belt-zone, becomes partially oxidized by high temperatures and blow-by gases, and forms insoluble resins which fuse to the hot piston. Overheating, as previously mentioned, raises piston temperatures which increase the accumulation rate of piston deposits.

A good maintenance history includes:

- Lubricating oil brand
- Lubricating oil viscosity
- Lubricating oil change intervals
- Full flow filter element and change interval
- Bypass filter element and change interval
- Air cleaner element and change interval.

In addition, the past history of the truck, with respect to filter plugging, use of lubricating oil and fuel additives, and the blending of lubricating oil and fuel **must** be obtained. This is useful information for correlations on why lubricating oil consumption problems occur at the rate they do today. Now that potential application problems and maintenance history have been reviewed, it is time to consider how the lubricating oil escapes from the engines.

External Leak Inspection

Examine the engine for external leakage before cleaning. Remember, while no single leak results in excessive lubricating oil consumption, several small leaks can make the difference. Check the crankcase breather tube for lubricating oil escaping. Some loss is normal.

Note all sources of external leakage so that they may be investigated and repaired. Measure the engine blow-by on a chassis dynamometer **only** if excessive lubricating oil loss appears to be occurring through the breather.

Internal Leak Inspection

Having eliminated or corrected external leakage problems, internal leakage inspection **must** be performed. Three contributors to internal leakage are turbocharger seals, valve guides, and air compressors.

To determine if the turbine side seals are leaking, remove the exhaust pipes and examine the turbine wheel for excessive deposits. Turbine seal leaks result in lubricating oil escaping from the engine into the exhaust.

Note : Wet exhaust ports do not always constitute lubricating oil consumption. Should an engine be run without turbocharger boost, it will overfuel and blow the excess fuel out of the exhaust.

To determine if the compressor side seals are leaking, remove the air crossover and examine it and the intake manifold. Excessive lubricating oil and/or lubricating oil residue indicates a leak.

To determine if the air compressor is responsible for internal leakage, check the truck air supply tanks and the air intake source of the air compressor for lubricating oil (intake manifold or air cleaner). If excess lubricating oil is present at either location, the air compressor is suspect.

Neither the turbocharger nor the air compressor individually can leak sufficient lubricating oil to result in excessive lubricating oil usage, but many small leaks together may result in excess lubricating oil usage and lubricating oil consumption. Repair the turbocharger and/or air compressor, if required.

Failure Analysis

Lubricating oil consumption in the combustion chamber is defined as lubricating oil leaking past the piston rings and burning. Causes of this loss of lubricating oil are cylinder liners, pistons, piston rings, and piston deposits. These will be considered separately from a failure analysis standpoint.

Piston and Piston Deposit Analysis

Piston top land and top ring groove deposits have been shown to cause the loss of lubricating oil control when they start to polish and rake the cylinder liners. Excessive top land and top ring groove deposits in themselves do **not** constitute a defect in material or workmanship. A piston exhibiting top land deposits which resulted in a loss of lubricating oil control is shown in Figure 2.

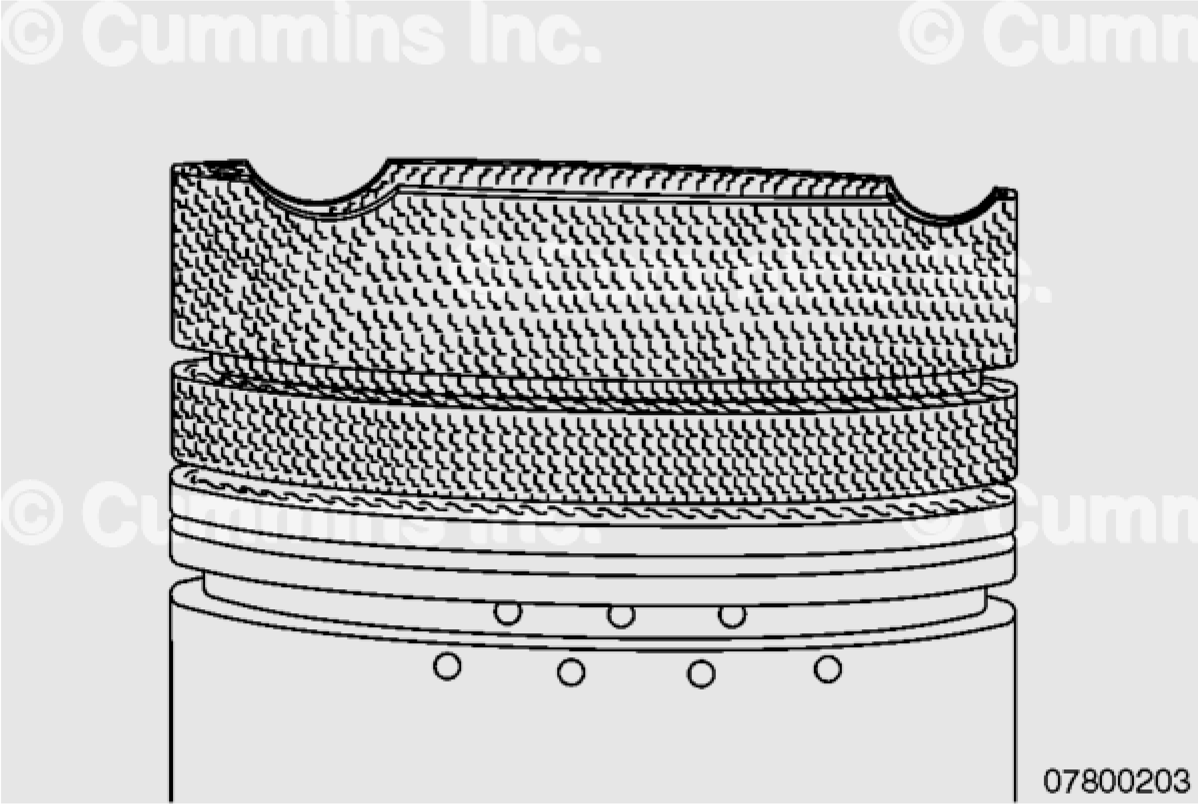

Figure 2:

A piston removed from an engine with lubricating oil consumption of 570 MPQ at 300,000 miles is shown in Figure 3.

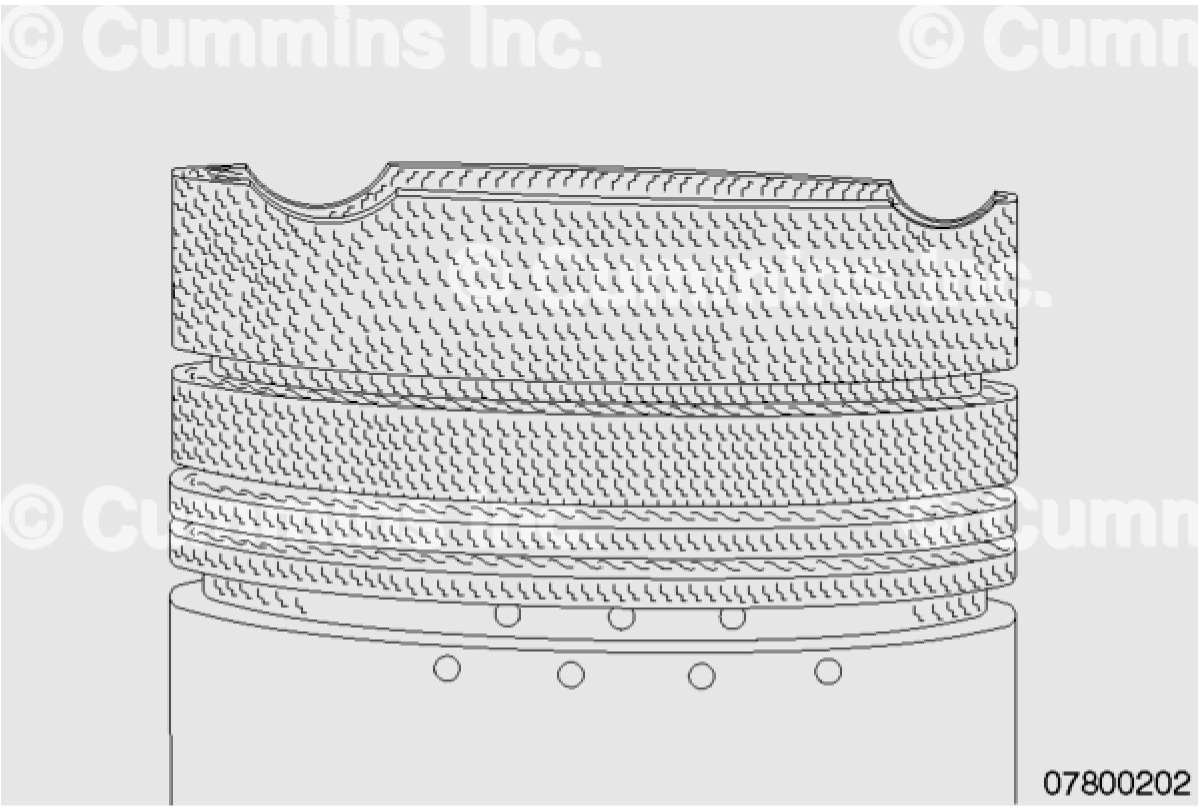

Figure 3:

This piston shown in Figure 3, does **not** exhibit heavy top land carbon deposits. In the laboratory, cleaning **only** the deposits from the piston top lands has been shown to restore baseline lubricating oil control without changing a single component.

Once the pistons are removed, the degree by which their deposits were affecting the lubricating oil consumption needs to be evaluated. The areas of the top land and in the groove, which have been polished, are areas where there was a loss of lubricating oil control. When performing the failure analysis, those areas need to be documented by estimating the percentages of fill for each piston and top groove. It takes **only** approximately 30 percent top land pack or 70 percent top groove fill for a piston to be the major contributor to lubricating oil consumption. See Figure 4.

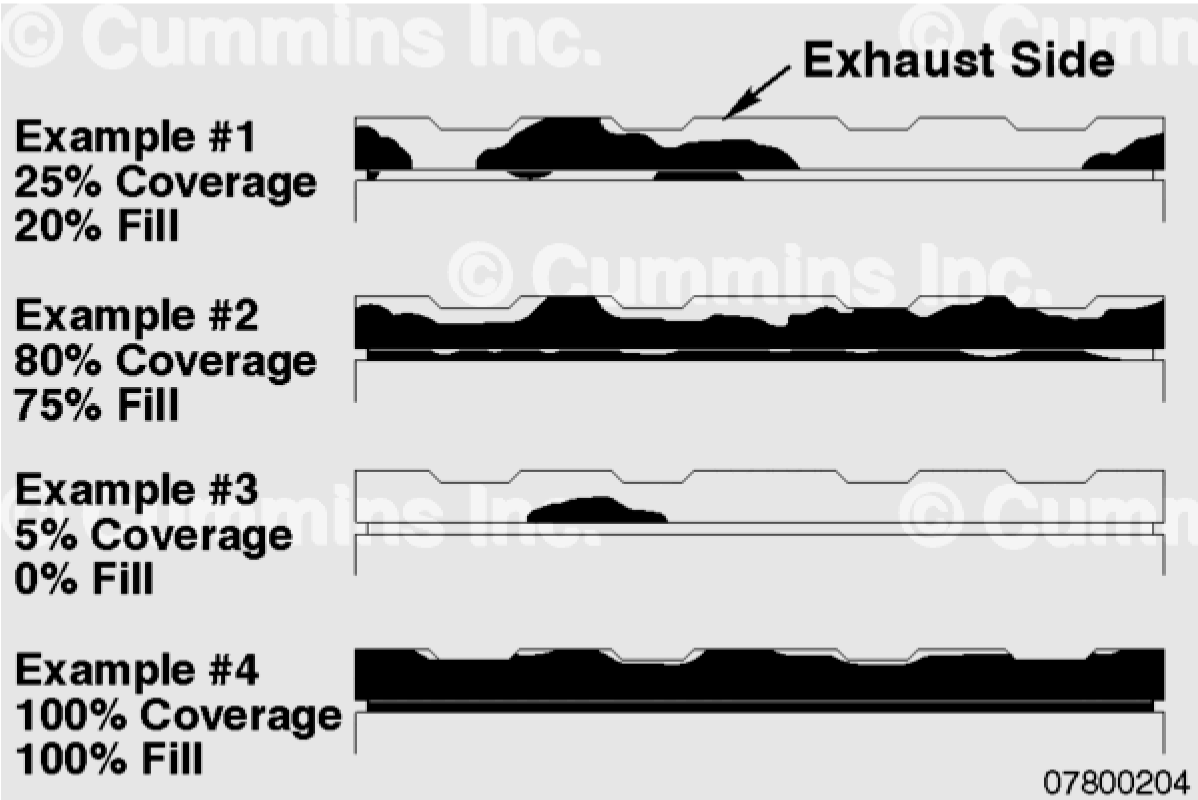

Figure 4: Piston Deposit Examples

Assembly Errors

Four piston ring assembly errors result in lubricating oil consumption:

- Upside-down intermediate rings
- Oil ring expander breakage or overlapping
- Missing rings
- Incorrect rings installed.

Upside-down intermediate rings are easily identified because the polished ring side, which seals on the lower piston groove, is **not** adjacent to the wear band on the ring face.

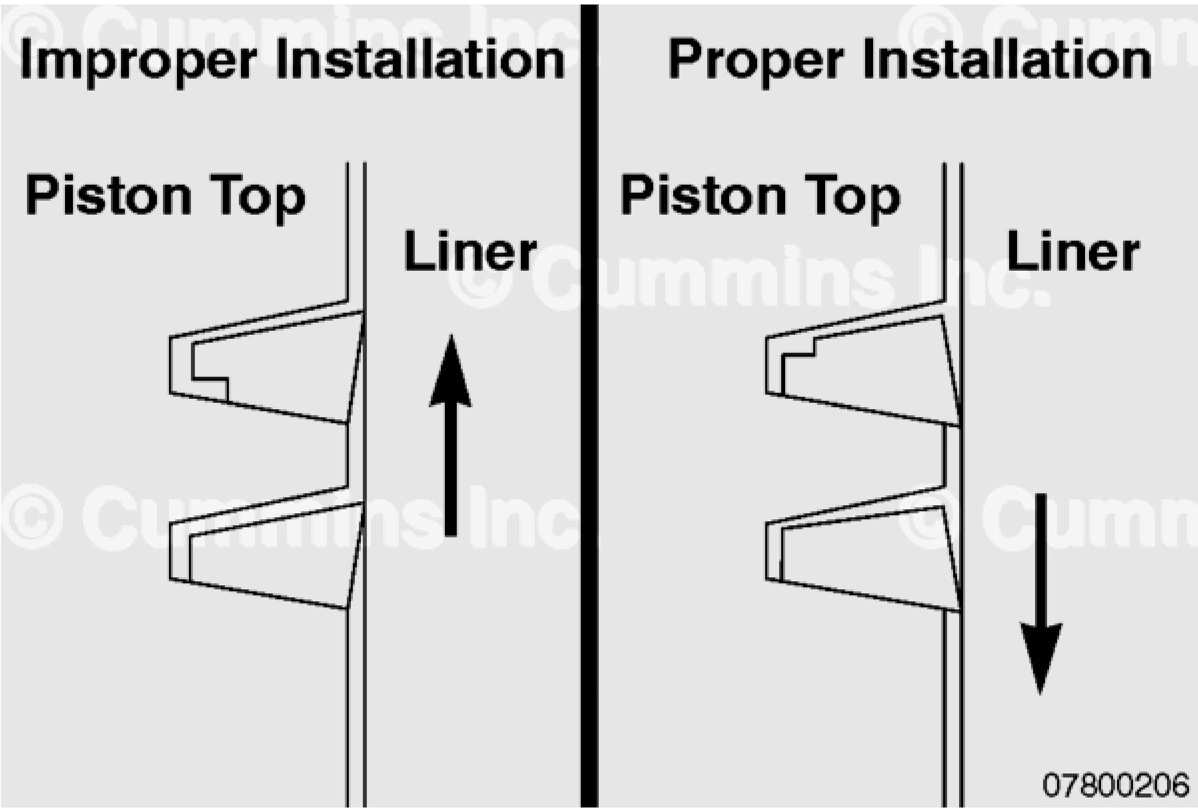

Figure 5: Intermediate Rings

Overlapped oil ring expanders may be identified by examining them before the oil ring is removed from the piston. Examination of the expander ends following removal will also show a characteristic polish pattern. See Figure 6.

Figure 6:

A peculiar seating pattern in the back of the piston oil ring groove also gives evidence of an overlapped expander.

Scuffing and Scoring

Scored or scuffed liners fall into two categories. The first being malfunctions below 15,000 miles usually caused by:

- Improper parts clearances
- Improper engine tests run-in.

The second category covers malfunctions above 15,000 miles caused by:

- Broken piston rings
- Abrasive wear
- Overheating
- Chrome flaking off the top ring.

Scuffed cylinder liners are easily identified by straw colored vertical scores as well as by an associated scuffed piston ring.

Many times the liner damage will be more severe and noticeable than piston ring damage, particularly early in the scuffing process when the lubricating oil control is initially lost. Liner scores from broken piston rings are characterized by a deep groove.

Piston problems which result in lubricating oil consumption are primarily ring groove wear caused by:

- Broken piston rings
- Abrasive wear
- Piston scoring
- Piston scuffing.

Piston scoring and scuffing may again be categorized as below and above 15,000 mile malfunctions due to the same reasons previously mentioned under scuffed and scored liners.

Piston ring defects which result in lubricating oil consumption are:

- Broken piston rings
- Incorrectly assembled piston rings
- Piston rings worn by abrasives
- Scuffed piston rings due to overheating
- Chrome flaking off the top ring.

Broken piston rings usually are caused by misassembly, over-expansion (when the break is 180 degrees to the end gap), and defects in the piston ring material. Ring side clearances which are excessive may also result in broken rings.

Abrasive Wear

Piston rings worn by abrasives are a controversial subject. The sources of the abrasives are well known: inadequate air filtration, inadequate lubricating oil filtration caused by filter plugging, and abrasives from dirty rebuilds or other external sources. The controversy involves the degree of wear to the rings. It is the most significant factor which should be noted. A full-face contact wear band on the top ring with the chrome plating worn through at the ring and gaps, and the first intermediate ring worn nearly to full-face contact are usually signs that the malfunction was caused by abrasives from inadequate air filtration. Oil rings worn wide and through the chrome plating, as well as significant wear of the lower intermediate rings, indicate abrasive wear from inadequate lubricating oil filtration. In all cases, the time since the engine was new or rebuilt should be taken into account.

Chaining, the snake-like marks observed on the lower ring side face, also indicates that abrasives were present in the engine. These marks are produced by a single large abrasive particle trapped between the moving ring and the stationary piston ring groove.

Liners worn by abrasives are very rare unless large amounts of dirt are involved. Should such wear patterns be found, inspect for gross inadequacies in the air and lubricating oil filtration systems. The records should be checked to verify the type of shop which last rebuilt the engine as built-in dirt or unauthorized liner honing abrasive may have been the source.

Liner wear cases are rare since the abrasive particles usually embed in the softer cast iron liner instead of taking seat in the harder piston rings (see Figure 7).

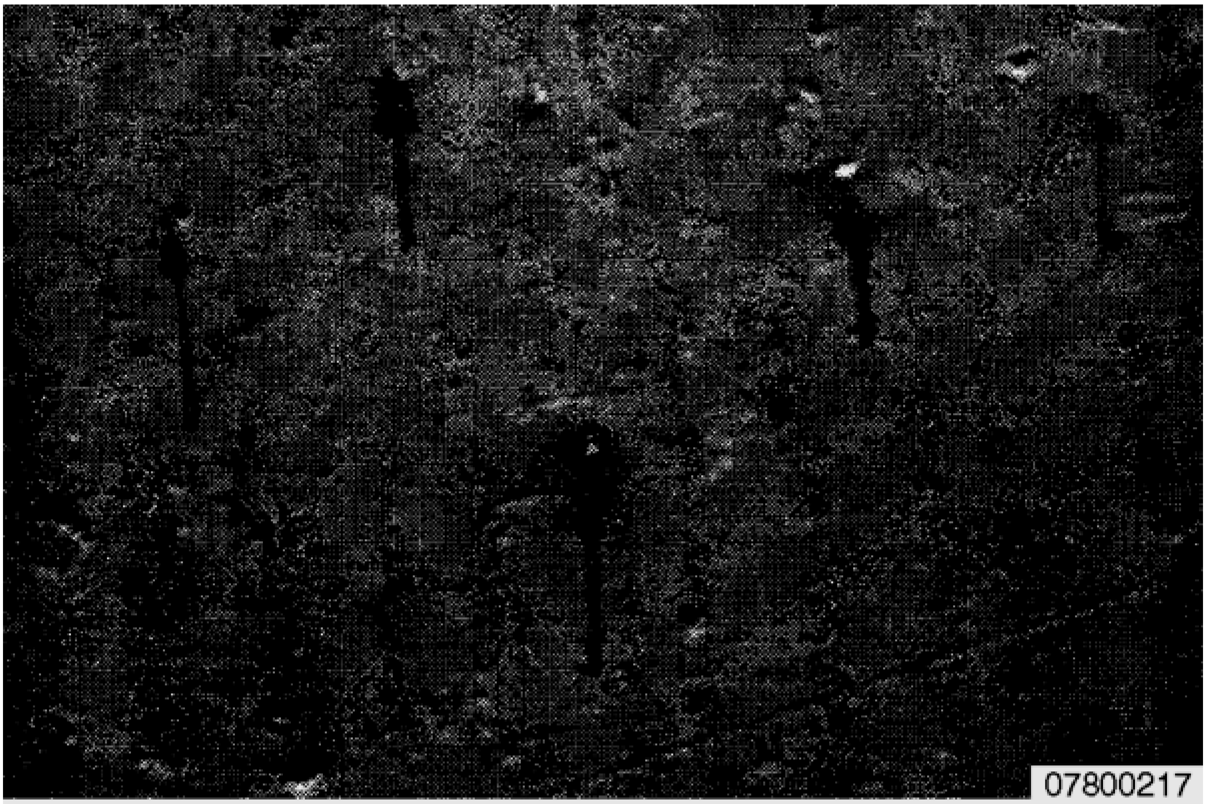

Figure 7:

Most wear from abrasives occurs to the moving piston rings which travel over the embedded abrasive particles. This was discussed earlier when piston ring wear was considered. Liner wear occurs **only** when extremely hard abrasives embed in the chromium plated top ring or when excessive dirt passes through the engine. In either case, liner wear in the ring travel area occurs.

Overheating

The causes of overheating are:

- Coolant loss
- Excessively high coolant temperatures
- Overfueling
- Improper cold starting procedures.

Chrome Flaking

Chrome flaking is the flaking off of the chrome plating from the top compression ring, usually initiating at the top edge of the ring. It occurs in both on-highway and off-highway applications.

Possible causes for chrome flaking are:

- Abrasives which wear the chrome thin, causing it to flake
- High sulfur fuels which attack the chrome plating and destroy the bond holding the plating to the iron base
- Poor initial chrome plating bond
- High temperatures which thermally destroy the chrome bond to the iron base.

Rebuild Procedures

Special instructions are required when rebuilding because of excessive lubricating oil consumption:

1. Reuse pistons, except those which are damaged or exhibit worn piston ring grooves.
2. Undamaged liners should also be reused, except for liners worn from inadequate air filtration problems. The liners will exhibit bright polish over the entire ring travel areas.
3. Cylinder liners and pistons to be reused should be cleaned by washing as described in the engine service manual and then inspected.
4. Extreme care **must** be exercised when cleaning the piston ring grooves so as **not** to distort them.

In Summary

1. Today's hardware is the best available with present technology.
2. A lubricating oil consumption complaint is a problem which **must** be evaluated on an engine-to-engine basis and **NOT** categorized or generalized. Today's challenge is to accurately determine the primary contributor(s) to the lubricating oil usage problems.
3. Evaluation of lubricating oil consumption complaints requires analysis of the application of the engine, engine maintenance history, external leaks, and internal engine components, in that order.
4. If external leaks or internal leaks are observed, repair them and evaluate the lubricating oil consumption for 5,000 miles or 30 days.
5. If external or internal leaks are **not** present, and the engine was **not** overfilled with lubricating oil, disassembly and inspection of the internal parts will be required. Closely evaluate air intake systems prior to disassembly for help in subsequent failure analysis.
6. By failure analysis, establish which of the following internal component problems caused extensive lubricating oil consumption:
7. Top land and top piston ring groove carbon deposits
8. Piston rings assembled incorrectly
9. Scuffed or scored liners and piston rings
10. Scuffed or scored pistons
11. Broken piston rings
12. Piston rings worn due to abrasives
13. Worn liners from abrasives
14. Worn piston ring grooves
15. Chrome flaking off top piston ring.
16. Assign the repair cost to the party responsible.
17. Repair the engine, reusing pistons and cylinder liners if they are **not** defective.

Lubricating Oil Consumption Report - 1

- Customer Name:____
- Engine Model:____
- Engine Serial Number:_______________________________________
- Vehicle Make/Model:____
- Distributor/Director:____
- Miles/Hours:____
- CPL Number:____
- Date:____

Check List Prior to Disassembly

1. Review Application
2. Check lubricating oil level; Too Full:_____, Correct:_____, Too Low:_____
3. Drain lubricating oil and refill to check dipstick markings; Overfilled by: _____qts, Underfilled by:_____ qts

4. History of overheating; Yes:_____, No:_____
5. History of overcooling; Yes:_____, No:_____
6. Air intake system: If applicable, follow procedure 010-024 in Section 10. See corresponding Service Manual.
7. Properly installed; Yes:_____, No:_____
8. New parts-ducting, etc.; Yes:_____, No:_____
9. Leaks in air intake system; Yes:_____, No:_____
10. All hose clamps tight; Yes:_____, No:_____
11. Any fabricated parts in air intake system; Yes:_____, No:_____
12. Review of Maintenance History

A. Previous malfunctions
B. Fuel in lubricating oil; Yes:_____, No:_____
C. Coolant in lubricating oil; Yes:_____, No:_____
D. Foreign material in lubricating oil; Yes:_____, No:_____
E. Turbocharger seals leaking; Yes:_____, No:_____
F. Injector problems; Yes:_____, No:_____
G. Lubricating Oil
H. Brand:______________
I. Viscosity:____________
J. Change interval:______________miles/hours
K. Bypass lubricating oil filter
L. Model:______________
M. Element:______________
N. Change interval:______________miles/hours
O. Full-flow filter
P. Element:______________
Q. Change interval:______________miles/hours
R. History of plugging; Yes:_____, No:_____
S. Air Cleaner
T. Make and Model:______________
U. Element:______________
V. Change interval:______________miles/hours
W. Ever use a reconditioned element; Yes:_____, No:_____

1. Check for External Leaks
2. Leaks from cracked or porous castings, related gaskets, and shaft seals
3. Cylinder block; Yes:_____, No:_____
4. Cylinder head(s); Yes:_____, No:_____
5. Turbocharger housing; Yes:_____, No:______
6. Gear covers; Yes:_____, No:_____
7. Valve covers; Yes:_____, No:_____
8. Rock box housing; Yes:_____, No:_____
9. Cambox housing; Yes:_____, No:_____
10. Crankshaft seals; Yes:_____, No:_____
11. Accessory drive seals; Yes:_____, No:_____
12. External lubricating oil lines and fittings; Yes:_____, No:_____

13. Lubricating oil filters; Yes:_____, No:_____
14. Lubricating oil lost from breather (road draft tube); Yes:_____, No:_____

Note : Steam clean only if necessary to determine exact locations of multiple leaks.

1. Check for Internal Leaks
2. Leaks by internal seals into combustion chambers or into the exhaust
3. Turbocharger turbine side seals; Yes:_____, No:_____
4. Turbocharger compressor side seals; Yes:_____, No:_____
5. Air compressor-excessive lubricating oil in air tanks; Yes:_____, No:_____
6. 5. **Only** after all above checks are done, leaks corrected, and multi-viscosity lubricating oil used, if lubricating oil consumption still persists, disassemble the engine to determine what has failed in the cylinders to cause the lubricating oil consumption.

Signed:____

Lubrication Oil Consumption Report - 2

- Customer Name:____
- Engine Model:____
- Engine Serial Number:____
- Vehicle Make/Model:____
- Distributor/Director:____
- Miles/Hours:____
- CPL Number:____
- Date:____

Failure Analysis Check List After Disassembly

Cylinder Liner Analysis

- Polish above top ring reversal from piston deposits; Yes:_____, No:_____
- Cylinder numbers and exhaust or intake side___________________; Yes:_____, No:_____
- Scuffed or scored liners cylinder numbers______________________; Yes:_____, No:_____
- Worn or deeply scratched liners cylinder numbers___________________________; Yes:_____, No:_____

Piston Analysis

- Piston deposits on top land and/or top groove
- Piston Number 1; Percent Land Coverage:_____, Percent Groove Fill:_____
- Piston Number 2; Percent Land Coverage:_____, Percent Groove Fill:_____
- Piston Number 3; Percent Land Coverage:_____, Percent Groove Fill:_____
- Piston Number 4; Percent Land Coverage:_____, Percent Groove Fill:_____
- Piston Number 5; Percent Land Coverage:_____, Percent Groove Fill:_____
- Piston Number 6; Percent Land Coverage:_____, Percent Groove Fill:_____

- (See examples in “Technical Overview of Oil Consumption” Book)
- Piston Number 1 Ring Groove Wear
- Top Ring - Out of Specification; Yes:_____, No:_____
- 1st Intermediate - Out of Specification; Yes:_____, No:_____
- 2nd Intermediate - Out of Specification; Yes:_____, No:_____
- Piston Number 2 Ring Groove Wear
- Top Ring - Out of Specification; Yes:_____, No:_____
- 1st Intermediate - Out of Specification; Yes:_____, No:_____
- 2nd Intermediate - Out of Specification; Yes:_____, No:_____
- Oil Ring - Out of Specification; Yes:_____, No:_____
- Piston Number 3 Ring Groove Wear
- Top Ring - Out of Specification; Yes:_____, No:_____
- 1st Intermediate - Out of Specification; Yes:_____, No:_____
- 2nd Intermediate - Out of Specification; Yes:_____, No:_____
- Oil Ring - Out of Specification; Yes:_____, No:_____
- Piston Number 4 Ring Groove Wear
- Top Ring - Out of Specification; Yes:_____, No:_____
- 1st Intermediate - Out of Specification; Yes:_____, No:_____
- 2nd Intermediate - Out of Specification; Yes:_____, No:_____
- Oil Ring - Out of Specification; Yes:_____, No:_____
- Piston Number 5 Ring Groove Wear
- Top Ring - Out of Specification; Yes:_____, No:_____
- 1st Intermediate - Out of Specification; Yes:_____, No:_____
- 2nd Intermediate - Out of Specification; Yes:_____, No:_____
- Oil Ring - Out of Specification; Yes:_____, No:_____
- Piston Number 6 Ring Groove Wear
- Top Ring - Out of Specification; Yes:_____, No:_____
- 1st Intermediate - Out of Specification; Yes:_____, No:_____
- 2nd Intermediate - Out of Specification; Yes:_____, No:_____
- Oil Ring - Out of Specification; Yes:_____, No:_____
- Scuffed or scored pistons
- Piston Number 1:_____
- Piston Number 2:_____
- Piston Number 3:_____
- Piston Number 4:_____
- Piston Number 5:_____
- Piston Number 6:_____

Piston Ring Analysis

- Upside down intermediate rings; Yes:_____, No:_____
- Cylinder numbers and ring locations:________________________________
- Overlapped oil ring expanders; Yes:_____, No:_____
- Cylinder numbers:________________________________
- Broken piston rings; Yes:_____, No:_____
- Cylinder numbers and ring type:________________________________
- Scuffed piston rings; Yes:_____, No:_____

- Cylinder numbers and ring type:______________________________
- Chrome flaking of top rings; Yes:_____, No:_____
- Cylinder numbers:______________________________

Piston Ring Wear/Amount of Wear

- Piston Number 1
- Top ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 1st Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 2nd Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- Piston Number 2
- Top ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 1st Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 2nd Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- Piston Number 3
- Top ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 1st Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 2nd Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- Piston Number 4
- Top ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 1st Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 2nd Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- Piston Number 5
- Top ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 1st Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 2nd Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- Piston Number 6
- Top ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 1st Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____
- 2nd Inter. ring; 0:_____, 1/4:_____, 1/2:_____, 3/4:_____, Full Face:_____

Signed:____

Document History

Date	Details
xxxx-xx-xx	Module Created
2014-1-20	Change oil consumption limits for ISX to conform with warranty alert a1342
2014-9-29	Revert ISX12 G to original oil usage spec
2015-4-21	Added ISG, ISG11, and QSG12 to Table 1.
2015-6-18	Addition of QSK95

Date	Details
2015-7-14	Addition of ISV
2016-2-25	Table 1 - Updated engine name from V/VT-550 to VT-555. Updated values for V/VT-378.
2016-3-23	Table 1 missing ISG11, ISG12, and QSG12 content.
2019-8-16	Added ISZ/QSZ to Table 1.
2020-6-25	Added ISB6.7 G/ B6.7N and ISL G/L9N to Table 1 and 2.
2022-11-7	Corrections made to Tables 1 and 2.
2023-1-25	Removed ISX15 in row for ISX/QSX and replace limits with 'Reference the ISX15 oil consumption evaluator under the "Service Tools" tab in QuickServe® Online'
2023-3-2	Updated Check List Prior to Disassembly section number 6.
2025-10-10	Added B6.7O to tables 1 and 2 and added B6.7O metadata to service bulletin.

Last Modified: 10-Oct-2025